

Essential Insights: Week 2.2



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The Layered Generative AI Stack

Seven-Layer Architecture

- **Hardware** – GPUs & AI accelerators
- **GPU Drivers** – CUDA ecosystem
- **ML Frameworks** – PyTorch, TensorFlow
- **Foundation Models** – LLMs & multimodal systems
- **Adaptation Techniques** – Fine-tuning, RAG
- **Inference & Serving** – vLLM, Triton
- **Applications** – User-facing tools (e.g., GitHub Copilot)

Key Insight: Success depends on how all layers work together, not on any single component.

Where Constraints Originate in the Stack

Constraints Appear Across Every Layer

- **Infrastructure:** High GPU & cloud costs (e.g., Klarna's experience)
- **Latency:** Slower responses with complex models under load
- **Governance:** Data privacy, compliance & risk (GDPR, audit trails)
- **Scalability:** Sudden traffic surges can degrade performance

Real-World Example: Klarna – 2.3 million AI conversations in one month, but needed hybrid human + AI model for quality and cost control.

Takeaway: Design the full stack holistically.

GPUs vs Specialised AI Accelerators

Hardware Choices Shape Cost, Speed & Flexibility

- **GPUs (NVIDIA H100, Blackwell B200):**
 - Dominant for training & general use
 - Strong ecosystem (CUDA) → developer preference & lock-in
 - High performance but expensive
- **Specialised Accelerators** (Google TPU, custom chips):
 - Optimised for inference & specific workloads
 - Better energy efficiency & lower cost per task
 - Less flexible outside their ecosystem

Business Decision: Balance flexibility (GPUs) vs efficiency (accelerators) based on workload.

Energy Bottlenecks & AI Sovereignty

Energy is the New Limiting Factor

- AI data centres consume ~30 GW globally (2026) – comparable to the Netherlands
- Training one frontier model can require 100–130 MW continuously
- Major companies securing nuclear & renewable energy (e.g., Microsoft + Three Mile Island)

Additional Challenges:

- Carbon emissions & sustainability regulations
- Grid constraints in growing markets (India, Middle East)
- AI sovereignty – national efforts to control chips, data centres & supply chains

Strategic Implication: Energy strategy is now core to AI strategy.

Hyperscalers – The Scaling Layer

Cloud Providers Enable Enterprise Scale

- **Role of Hyperscalers (Azure, AWS, Google Cloud):**
 - Elastic GPU clusters that scale automatically with demand
 - High-speed interconnects (NVLink) for large model training & inference
 - Load balancing & tiered routing (simple vs complex queries)
 - Orchestration with Kubernetes for resilience & failover

Real Example: GitHub Copilot – serves tens of millions of developers with sub-second responses during global peaks.

Key Benefit: Organisations focus on value creation, not building data centres.

Foundation Models Layer

The Strategic Core of Generative AI

- **Foundation Models:** Large-scale pretrained systems (GPT-5, Claude Opus 4, Gemini 2.5 Pro, Llama 4)
- Trained on trillions of tokens using next-token prediction
- Increasingly **multimodal** – handle text, images, audio & video
- Long-context capabilities (up to millions of tokens)

Enterprise Applications:

- Ambient AI scribes in healthcare (e.g., Microsoft DAX, Nabla)
- Reduced clinician burnout and documentation time

Reality Check: Training costs run into tens or hundreds of millions – most organisations adapt existing models.

Adaptation Techniques

Making Foundation Models Business-Ready

- **Fine-tuning:** Adjust model with domain-specific data
- **Retrieval-Augmented Generation (RAG):** Connect model to internal documents/knowledge bases
- Reduces hallucinations and improves accuracy

Combined Use:

Most successful enterprise deployments combine foundation models with RAG + selective fine-tuning.

Serving Engines – vLLM & Triton

The Hidden Layer That Determines Real-World Performance

- **Serving Engine:** Manages requests, batching, resource allocation & response delivery
- Critical during traffic surges (e.g., airline disruption support)

Key Tools:

- **vLLM:** Memory-efficient serving for LLMs → higher throughput
- **NVIDIA Triton:** Unified serving for multiple model types

Impact Example: One bank increased document processing from 150 to 600 per second on the same hardware through better serving optimisation.

Takeaway: A great model with poor serving = slow & expensive system.

Quantisation & Cost Control

Reducing Inference Costs Without Major Quality Loss

- **Problem:** Inference can account for 60–80% of total AI costs
- **Solution – Quantisation:** Reduce precision of model weights (16-bit → 8-bit or 4-bit)

Benefits:

- Lower GPU memory usage
- Higher throughput (more queries per GPU)
- Significant cost savings

Real-World Result: European regtech firm doubled summaries processed per GPU hour with 8-bit quantisation, while maintaining usable accuracy.

Strategy: Use lower-precision models for high-volume routine tasks; keep full precision for complex reasoning.

Full Stack Summary – Key Layers & Decisions

Generative AI Success = Full-Stack Optimisation

Layer	Key Decisions	Main Constraints
Hardware	GPU vs Accelerator	Cost & Energy
Cloud / Hyperscalers	Scaling & orchestration	Latency & Reliability
Foundation Models	Selection & multimodal capability	Capability & Cost
Serving & Inference	vLLM, Triton, batching	Throughput & Speed
Adaptation	RAG + Fine-tuning	Accuracy & Hallucinations
Applications	User experience & integration	Adoption & Value

Core Principle: Optimise across the entire stack, not in isolation.

Final Takeaways

- Generative AI is a **complex, layered infrastructure stack**, not just an app
- Constraints (cost, latency, governance, energy) emerge across all layers
- Hardware & energy decisions are now strategic, not purely technical
- Hyperscalers provide the scaling backbone
- Foundation models + adaptation techniques (RAG, fine-tuning) deliver business value
- Serving engines and quantisation are critical for performance and cost control

Treat generative AI as a full-stack capability. Leaders who understand and optimise the entire stack will achieve scalable, reliable, and responsible AI deployment.